

HDOC - Day 15 Activity

C Programming Assignment

Name-Arush Gusain

Sap id-590033489

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Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

1. Check whether all digits are even.
2. Check whether all digits are prime.

Algorithm

1. Start.
2. Input a positive integer number.
3. If number ≤ 0 , display an invalid-input message and stop.
4. Display the menu: Choice 1 - check all digits even; Choice 2 - check all digits prime.
5. Input the user's choice and copy number into temp.
6. Use a switch(choice) statement.
7. For Choice 1, repeatedly obtain the last digit using $\text{digit} = \text{temp} \% 10$. If any digit is odd ($\text{digit} \% 2 \neq 0$), mark the condition false. Otherwise remove the last digit using $\text{temp} = \text{temp} / 10$. Display the result.
8. For Choice 2, repeatedly obtain each digit. A digit is prime only if it is 2, 3, 5, or 7. If any other digit occurs, mark the condition false. Display the result.
9. For any other choice, display Invalid choice.
10. Stop.

Test Case Table

S. No.	Input(number)	Choice	Expected output	Result
1.	24680	1	All digits are even	Pass
2.	12345	1	All digits are not even	Pass
3.	2357	2	All digits are prime	Pass
4.	2379	2	All digits are not prime	Pass
5.	2468	3	Invalid choice	Pass

C Program

```
#include <stdio.h>
int main(void) {
    long long number, temp;
    int choice, digit;
    int allEven = 1, allPrime = 1;
    printf("Enter a positive integer: ");
    scanf("%lld", &number);
    if (number <= 0) {
        printf("Invalid input. Please enter a positive integer.\n");
        return 0;
    }
    printf("\n1. Check whether all digits are even\n");
    printf("2. Check whether all digits are prime\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    temp = number;
    switch (choice) {
        case 1:
            while (temp > 0) {
                digit = temp % 10;
                if (digit % 2 != 0) {
                    allEven = 0;
                    break;
                }
                temp /= 10;
            }
            if (allEven)
                printf("All digits are even\n");
```

```
else
printf("All digits are not even\n");
break;
case 2:
while (temp > 0) {
digit = temp % 10;
if (!(digit == 2 || digit == 3 || digit == 5 || digit == 7)) {
allPrime = 0;
break;
}
temp /= 10;
}
if (allPrime)
printf("All digits are prime\n");
else
printf("All digits are not prime\n");
break;
default:
printf("Invalid choice\n");
}
return 0;
}
```

Compilation , Testing & Output

```
Se... Simple Text Editor Help
(kali@kali)~$ ./a.out
Enter a positive integer: 24680

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are even

(kali@kali)~$ ./a.out
Enter a positive integer: 12345

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are not even

(kali@kali)~$ ./a.out
Enter a positive integer: 2357

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are prime

(kali@kali)~$ ./a.out
Enter a positive integer: 2379

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are not prime

(kali@kali)~$ ./a.out
Enter a positive integer: 2468

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 3
Invalid choice
```